	The total number of marks available for this examination paper is 80. It contributes 40% to AS Physics and 20% to the Advanced GCE in Physics.
Calculator	It is recommended that students have access to a scientific calculator for this paper.
Formulae sheet	Students will be provided with the formulae sheet shown in *Appendix 8*: *Formulae*. Any other physics formulae that are required will be stated in the question paper.

1.3 Mechanics

This topic leads on from the Key Stage 4 programme of study and covers rectilinear motion, forces, energy and power. It may be studied using applications that relate to mechanics, for example, sports.

Students will be assessed on their ability to:	**Suggested experiments**
1 use the equations for uniformly accelerated motion in one dimension: $v = u + at$ $s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$	
2 demonstrate an understanding of how ICT can be used to collect data for, and display, displacement/time and velocity/time graphs for uniformly accelerated motion and compare this with traditional methods in terms of reliability and validity of data	Determine speed and acceleration, for example use light gates
3 identify and use the physical quantities derived from the slopes and areas of displacement/time and velocity/time graphs, including cases of non-uniform acceleration	
4 investigate, using primary data, recognise and make use of the independence of vertical and horizontal motion of a projectile moving freely under gravity	Strobe photography or video camera to analyse motion
5 distinguish between scalar and vector quantities and give examples of each	
6 resolve a vector into two components at right angles to each other by drawing and by calculation	
7 combine two coplanar vectors at any angle to each other by drawing, and at right angles to each other by calculation	

Students will be assessed on their ability to:	Suggested experiments
8 draw and interpret free-body force diagrams to represent forces on a particle or on an extended but rigid body, using the concept of *centre of gravity* of an extended body	Find the centre of gravity of an irregular rod
9 investigate, by collecting primary data, and use $\Sigma F = ma$ in situations where m is constant (Newton's first law of motion ($a = 0$) and second law of motion)	Use an air track to investigate factors affecting acceleration
10 use the expressions for gravitational field strength $g = F/m$ and weight $W = mg$	Measure g using, for example, light gates. Estimate, and then measure, the weight of familiar objects
11 identify pairs of forces constituting an interaction between two bodies (Newton's third law of motion)	
12 use the relationship $E_k = ½ mv^2$ for the kinetic energy of a body	
13 use the relationship $\Delta E_{grav} = mg\Delta h$ for the gravitational potential energy transferred near the Earth's surface	
14 investigate and apply the principle of conservation of energy including use of work done, gravitational potential energy and kinetic energy	Use, for example, light gates to investigate the speed of a falling object
15 use the expression for work $\Delta W = F\Delta s$ including calculations when the force is not along the line of motion	
16 understand some applications of mechanics, for example to safety or to sports	